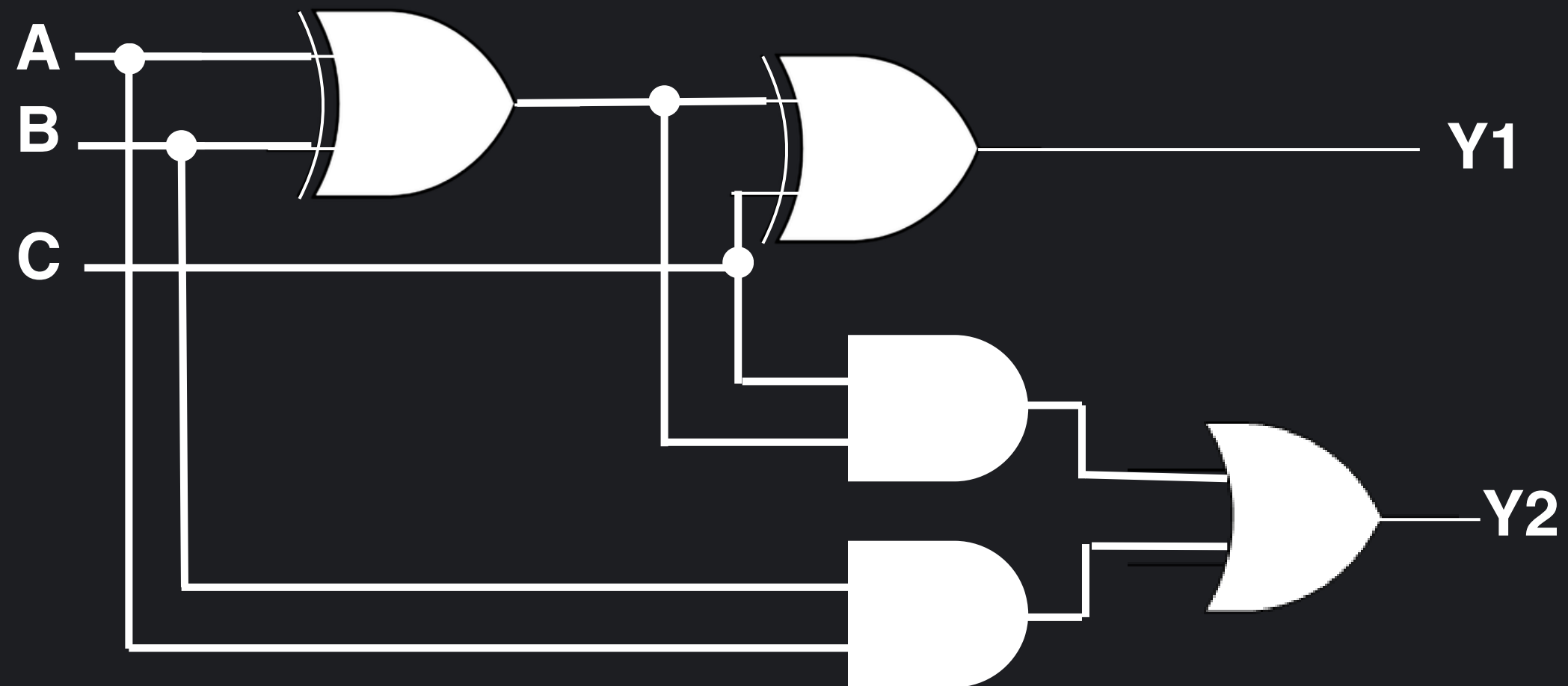


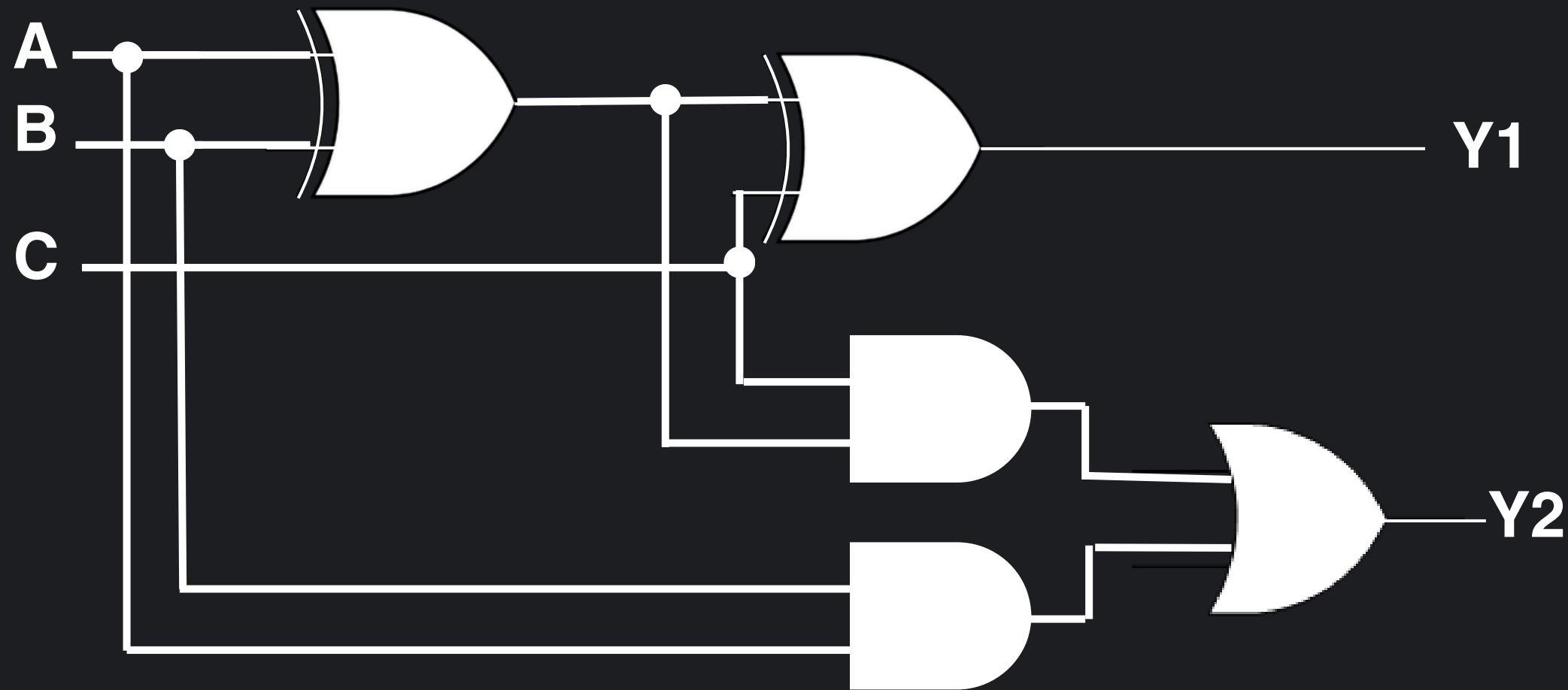
FULL ADDER



What is a FULL ADDER?

A Full Adder is a basic combinational logic element that performs the arithmetic addition of three binary bits. It produces two outputs, Sum and Carry, based on its three inputs which consist of two operand bits and a Carry-in from a previous stage.

Truth Table



A	B	C	Y1	Y2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Function of the FULL ADDER

A Full Adder performs a logical arithmetic addition on its three binary inputs. The main function of a Full Adder is to serve as a basic arithmetic element and produce an output signal (Sum and Carry Out) according to this rule:

Output Sum = 1 when the total of inputs (A, B, Cin) is 1 or 3.

Output Sum = 0 when the total of inputs is 0 or 2.

Output Carry Out = 1 when at least two inputs are 1.

Output Carry Out = 0 when fewer than two inputs are 1.